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Design, testing and control of a magnetorheological actuator for assistive knee braces

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Abstract
This paper is aimed at developing a smart actuator for assistive knee braces to provide assistance to disabled or elderly people with mobility problems. A magnetorheological (MR) actuator is developed to be used in assistive knee braces to provide controllable torque. The MR actuator can work as a brake or a clutch. When active torque is needed, the DC motor works and the MR actuator functions as a clutch to transfer the torque generated by the motor to the leg; when passive torque is desired, the DC motor is turned off and the MR actuator functions as a brake to provide controllable passive torque. The prototype of the developed MR actuator is fabricated and experiments are carried out to investigate the characteristics of the MR actuator. The results show that the MR actuator is able to provide sufficient torque needed for normal human activities. Adaptive control is proposed for controlling the MR actuator. Anti-windup strategy is used to achieve better control performance. Experiments on the MR actuator under control are also performed to study the torque tracking ability of the system.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

The percentage of aged persons in society is increasing, and their physical deterioration has become a socioeconomic problem in many countries, such as China, Japan, etc. In late 2005, the population of aged persons (older than 60 years) in China had reached 143 million. In this century, the aged tendency of the population will become more serious. It is estimated that the percentage of aged population would reach 30% in China between 2030 and 2050. Elderly people with weak muscle strength may not be able to walk as frequently as before and would also lose their stability during walking. In turn, their muscles would be further deteriorated and they may become bedridden. The most effective means to avoid people from becoming bedridden is to provide a way for them to be able to maintain their physical activities.

Osteoarthritis (OA) is very prevalent in old people; it is about 60% in men and 70% in women after the age of 65 years [1]. Studies showed that adequate exercises would generate positive effects to OA patients. Some kinds of devices were developed to provide exercises; however, these devices are usually for hospital or home use and cannot automatically provide assistance for people with mobility problems. Considering the large population base and the high percentage of elderly people, devices that can help people with weak muscle strength or OA will be in great need. Regarding this problem, research was conducted to develop exoskeletons that can provide assistance for elderly and disabled people, such as RoboKnee developed by Pratt *et al* [2], and the hybrid assistive limb (HAL) developed by Sankai *et al* [3, 4]. Although great progress has been made, these leg exoskeletons still have some limitations such as high power consumption, safety concerns, etc.

MR fluids comprise microscale ferromagnetic particulates dispersed in a carrier liquid. When a magnetic field is applied to this fluid, the dispersed magnetic particles will line up along the direction of the magnetic field. A shear stress is needed to disrupt this chain structure. The apparent yield stress increases

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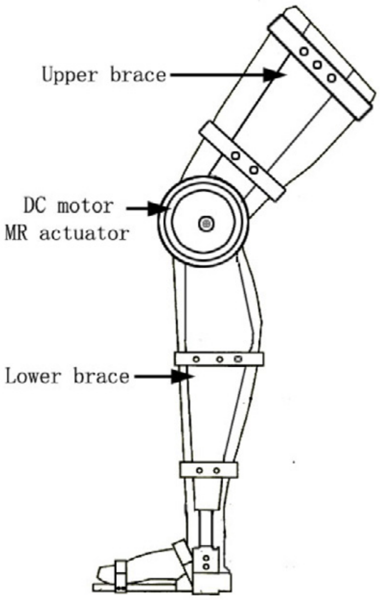
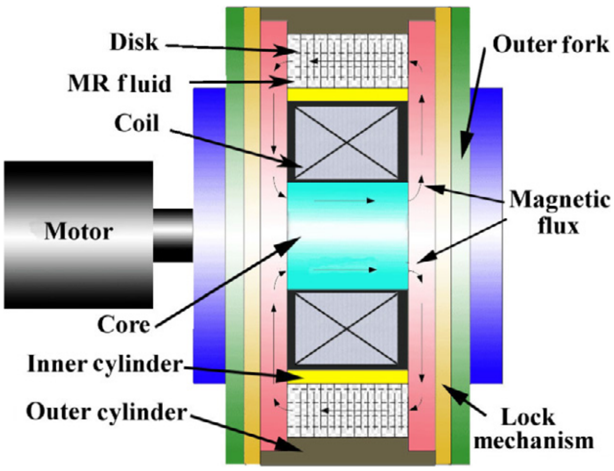


(a)



(b)

Figure 1. Configuration of (a) knee brace and (b) MR actuator.

as the magnetic field increases. As a smart fluid, an MR fluid has several advantages such as high yield stress, good stability and fast response time. These features make the MR fluid promising for various applications. A remarkable application of MR fluids is smart prosthetics, where the MR damper is used to provide controllable damping to the knee joint and thus make the prosthesis adaptable to the person while interacting with the environment. Powered by a lithium-ion battery, the system is able to operate for two days when fully charged [5]. the MR damper was also used in a rehabilitation device to provide controllable resistance with excellent force tracking performance [6–9].

Programmable aerobic exercise equipment that uses an MR brake as the controllable resistance element has been commercialized, where the MR brake can provide a maximum torque of more than 7 N m [10]. Zite *et al* used an MR brake in an orthopedic active knee brace to provide controllable

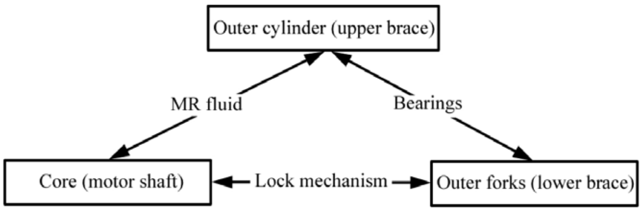


Figure 2. Motion relationship of MR actuator.

resistance [11, 12]. The maximum torque produced by the MR brake is 6 N m when the applied coil current is 4.0 A [12]. Herr and his associates developed a prosthetic knee with an MR brake. The MR brake can produce a torque of 0.5–40 N m [13, 14]. Li *et al* developed an MR brake used in a prosthetic ankle joint to make the user walk smoothly [15], but more work is needed to put this system into practical use. These MR brakes can only provide controllable passive force or torque.

Takesue *et al* developed an MR fluid actuator, which consists of an MR fluid clutch between input and output parts [16, 17]. This device has the benefits of safety and low inertia. However, the actuator cannot provide large torque (the maximum torque is about 5 N m), and it is a bit bulky (weight about 5 kg). Both MR actuator and motor should work together to provide assistive torque.

Since both active and passive torques are needed for assistive knee braces, it would be desirable if an MR actuator can combine the advantages of an MR brake (to produce controllable torque while consuming little power) and an MR clutch (to safely transfer torque from the motor to the leg). In this study, an MR actuator that can function as a brake or a clutch, as needed, will be developed and tested. Adaptive control is proposed to control the developed MR actuator, and the torque tracking ability of the MR actuator under control will also be studied.

2. Design and analysis of MR actuator

The assistive knee brace is shown in figure 1(a), where the MR actuator will provide assistive torque to help people with their mobility. To produce large torque and keep the MR actuator compact in size, the MR actuator utilizes a plurality of interspersed and alternating rotors and stators. The principle of the MR actuator is shown in figure 1(b). It consists of a DC motor, which provides active torque when needed, an outer cylinder, which is connected to the upper leg, MR fluids that are used in shear mode to produce or transfer torque, and an inner cylinder, which is bonded to the shaft. The outer discs (stators) are engaged in the outer cylinder; the inner discs (rotors) are engaged in the inner cylinder. MR fluids are filled between the outer and inner discs. The shaft is connected to the motor, and the motor is mounted on the lower brace. The inner cylinder (or the shaft) can be locked to the fork, which is connected to the lower brace.

The motion relationship of the MR actuator is shown in figure 2. For this actuator, there are three working conditions:

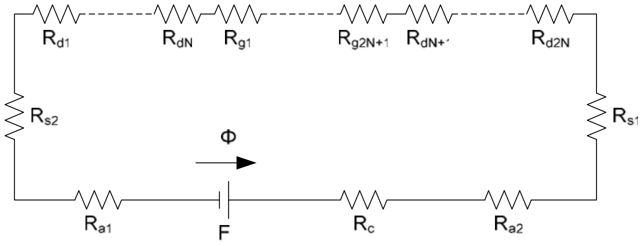


Figure 3. Equivalent magnetic circuit of MR actuator.

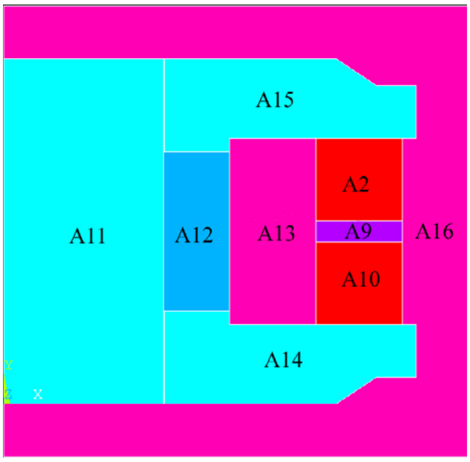


Figure 4. Magnetic loop of the MR actuator.

(1) The shaft is locked to the lower brace and the magnetic field is on. The MR device acts as a brake, which can provide controllable passive torque.
(2) The magnetic field is off. The MR device does not work and the knee joint can rotate freely.
(3) The shaft is unlocked to the lower brace and the magnetic field is on, while the motor is working. As the shaft is unlocked to the lower brace, the shaft can rotate relative to the lower leg, thus the MR device acts as a clutch to transfer the torque from the motor to the upper leg.

The main functional components are the alternately placed outer discs (or stators) and inner discs (or rotors). The inner discs can rotate relative to the outer discs as the shaft or the knee rotates. MR fluids fill the gaps between the outer and inner discs. As the shaft rotates and the magnetic field is on, the torque due to the shear effect will be generated between the outer and inner discs. This torque is controlled by the applied magnetic field.

The torque generated by the actuator under post-yield state can be calculated as follows:

$$T = 2N \int_{R_1}^{R_2} 2\tau\pi r^2\,\mathrm{d}r = 4N\pi\left[\frac{\tau_y(R_2^3 - R_1^3)}{3} + \frac{\mu\omega(R_2^4 - R_1^4)}{4w}\right] \qquad (1)$$

where N is the number of outer discs, R_1 and R_2 are the inner and outer radii of overlapping areas of the discs from the axis of rotation, τ_y is the yield stress, μ is the viscosity of the MR fluid,

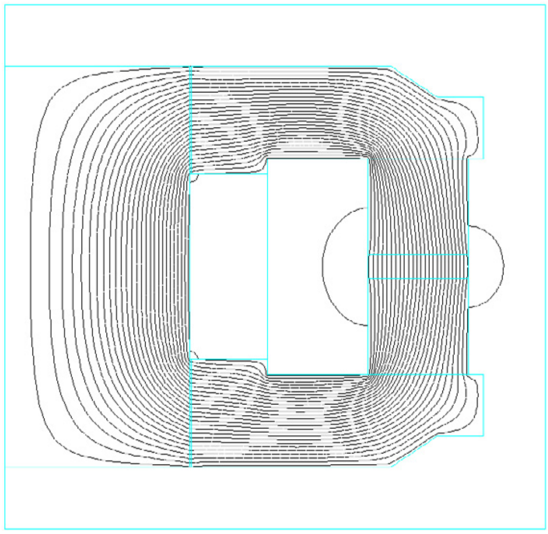


Figure 5. The magnetic flux contour at 1.5 A.

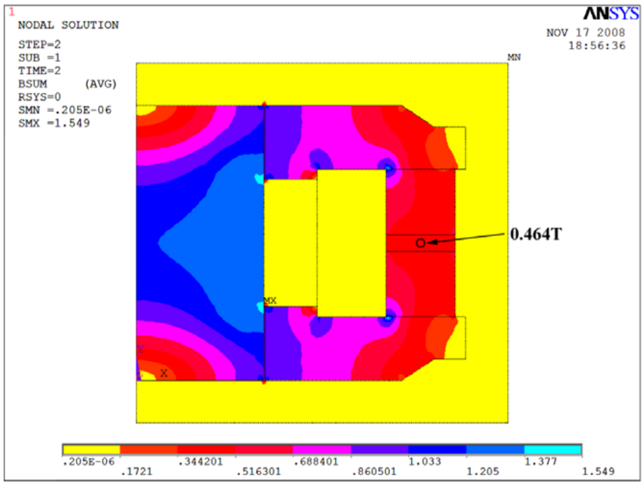


Figure 6. The magnetic flux density contour at 1.5 A.

ω is the angular velocity of the inner disc and w is the width of the gap between the adjacent outer disc and inner disc. The first term of (1) is produced by the yield stress and the second term is generated by the viscosity of the MR fluid.

For the MR actuator, the magnetic field applied to MR fluids determines the on-state torque. Therefore it is important to design an appropriate magnetic circuit. As shown in figure 1(b), the magnetic circuit is formed by the shaft core, side plates, the air gaps between the shaft core and side plates, discs and MR fluids. If the magnetic circuit is not saturated, the equation of the magnetic circuit is as follows:

$$nI = \sum_{i=1}^{m} R_i \Phi_i \qquad (2)$$

where I is the coil current, n is the number of coils, R_i is the magnetic reluctance, Φ_i is the magnetic flux and

$$R_i = \frac{l_i}{\mu_i S_i} \qquad (3)$$

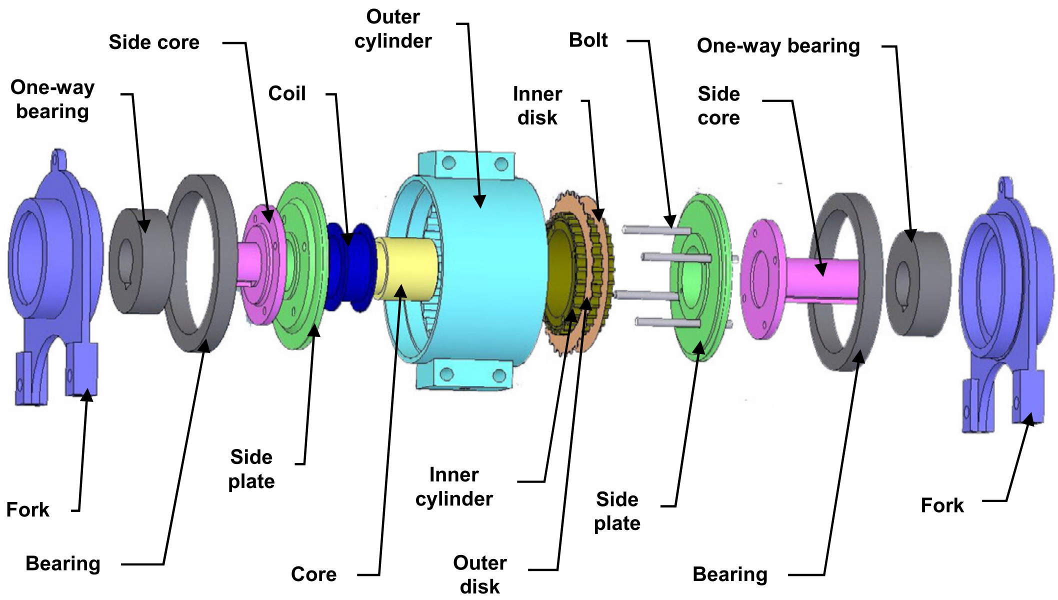


Figure 7. Exploded view of MR actuator.

where l_i is the length of the magnetic circuit component, S_i is the area of the cross section and μ_i is the magnetic permeability. Since the magnetic circuit components are connected in series, the magnetic induction is the same in the circuit, so let $\Phi = \Phi_i$. The equivalent magnetic circuit is shown in figure 3. It is assumed that all the discs are made of the same material with the same size; also the gaps between the discs are the same. Therefore, we can obtain

$$\Phi = \frac{nI}{\sum_{i=1}^{m} R_i} = \frac{nI}{R_c + 2R_s + 2R_a + 2NR_d + (2N+1)R_g} \tag{4}$$

where R_c, R_s, R_a, R_d and R_g are the magnetic reluctance of shaft core, side plate, air gap between the shaft core and side plate, disc and the gap between discs (filled with MR fluids), respectively.

The magnetic flux will not always increase with the coil current. When one component is saturated, the magnetic flux will not be increased further. Therefore

$$\Phi_{max} = \min(B_{cs}S_c, B_{ss}S_s, B_aS_a, B_{ds}S_d, B_{gs}S_g) \tag{5}$$

where B_{cs}, S_c, B_{ss}, S_s, B_a, S_a, B_{ds}, S_d, B_{gs} and S_g are the saturation magnetic flux density and area of shaft core, side plate, air gap, disc and MR fluid gap, respectively.

To improve the performance of the MR actuator, the finite element method (FEM) is used to analyze the magnetic circuit. From (4), we know that the magnetic flux will remain the same if we change the sequence of the magnetic circuit components. Therefore, we can simplify finite element analysis by putting the MR fluid gaps together.

The designed magnetic loop is shown in figure 4. A11 is the shaft core, A14 and A15 are the side plates, A12 is the coil, A2 and A10 are the discs (multiple discs connected), A9 is the MR fluid (all gaps are connected), A13 is the inner cylinder and A16 (including outer cylinder, side cores and air) is used to define the boundary condition in the FEM simulation.

The diameter and length of the shaft core is 24 mm and 26 mm, respectively. The diameter of the side plates is 64 mm and the thickness is 7 mm in the inner circle and 4 mm in the outer circle. The dimensions of the discs are $R_1 = 23.5$ mm and $R_2 = 30$ mm. The height of A9 is 1.6 mm. The air gap between the shaft core and side plate is 0.1 mm. The shaft core and side plates are made of pure iron; the material of the discs is silicon steel; the MR fluid is MRF-132DG from Lord Inc. The outer and inner cylinders as well as the side cores are made of aluminum alloy 6061 T. The relative magnetic permeability is 1 for non-magnetic materials (including aluminum alloy and air). The FEM simulation is carried out using ANSYS. A four-node element plane13 with 2D magnetic, thermal, electrical, piezoelectric and structural field capability is used in the modeling.

The magnetic flux contour and magnetic flux density at 1.5 A excitation current are shown in figures 5 and 6, respectively. The maximum magnetic flux density at the shaft core is 1.549 T, which is below the saturation threshold of the shaft core material (i.e. pure iron, whose saturated magnetic flux density is 1.8 T). The magnetic flux density in the MR fluid is between 0.344 and 0.516 T. Let $N = 20$, then from equation (1), the calculated torque generated by the MR actuator is about 27 N m. It was shown, [4], that the estimated torque for a knee joint during normal walking is smaller than 20 N m. The maximum torque generated by the actuator could be further increased by the following methods:

(1) Decrease the air gap between the shaft core and side plates. Since the magnetic circuit is not saturated, the magnetic flux is determined by (4). The magnetic permeability of air is much lower than other materials in the magnetic circuit. So decreasing R_a will decrease the magnetic resistance and thus increase the magnetic flux.
(2) Increase the number of total discs by reducing the thickness of the discs. As we can see from equation (1), while more discs are used, higher torque can be generated.

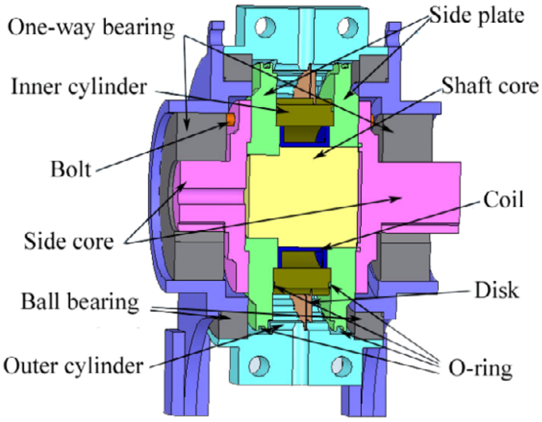


Figure 8. Sectional view of assembled MR actuator.

For normal walking, the knee power is mainly passive, which usually happens during the bending of the knee joint; active power occurs during the extending of the knee [18]. Accordingly, a prototype is designed and fabricated as shown in figures 7–9. A pair of one-way bearings are used to shift the MR fluids to function between brake and clutch. The lower brace is mechanically communicating with the shaft core through the one-way bearings. At the same time, the lower brace is communicating with the upper brace by normal ball bearings, as shown in figure 2. The one-way bearings can only rotate in the knee extending direction, i.e. the one-way bearings are locked in the bending direction and unlocked in the extending direction. If no magnetic field is applied to the MR fluids, the lower brace can freely rotate with respect to the upper brace. If a magnetic field is applied to the MR fluids and the knee joint is extending, the one-way bearings are unlocked and torque generated by the motor (which is mounted on the lower brace) can be passed to the upper brace through one-way bearings and MR fluids. If a magnetic field is applied to the MR fluids and the knee is bending, the one-way bearings are locked so that the MR device functions as a brake to provide passive torque.

3. Testing of MR actuator

Experiments are carried out in order to investigate the characteristics of the developed MR actuator. Figures 10 and 11 show the block diagram and photo of the experimental set-up. The shaft core of the MR actuator is rotated at a certain speed in one direction by an AC motor through a 10:1 gearbox. Here the MR actuator functions as a clutch. The torque is transmitted from the shaft core to the outer cylinder by applying the current to the coil. In order to measure the output torque of the MR actuator, a load cell (Model 208c03

(a)

(b)

Figure 9. Photos of the MR actuator: (a) components and (b) assembled actuator.

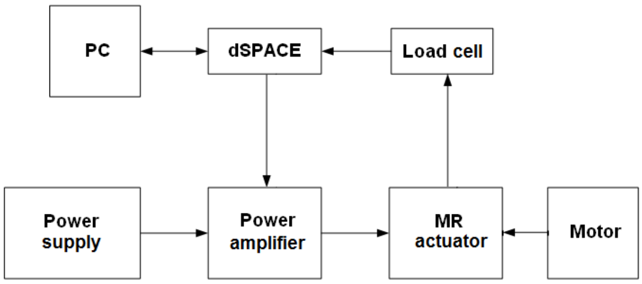


Figure 10. Block diagram of the experimental set-up.

from PCB Inc.) is mounted on the outer cylinder of the MR actuator, which is fixed on a vibration-free table.

The MR actuator rotates at a constant speed of 1.4 rad s^{-1} and the output torque is measured by the load cell under a step input coil current of 0.5, 1.0, 1.5 and 1.8 A, respectively. The experimental result while applying the current 1.0 A is shown in figure 12. It can be seen from the figure that the response time (the time needed for the output of the MR actuator to reach 90% of its steady output under a step input coil current) of the MR actuator is about 0.1 s (for comparison, the reaction time of a normal human is between 0.15 and 0.40 s). Since the response time of the MR fluid itself is several milliseconds, faster response could be achieved by using better magnetic materials (such as materials with high electric resistivity to reduce eddy currents). High frequency noise appears in the measured torque. The noise can be filtered out by a low-pass filter, as shown in figure 13, where a low-pass Butterworth filter is used. The filter order is set to 1 and the cutoff frequency is 100 rad s^{-1}.

Figure 14 shows the relationship between average torque and applied coil current. We can see that the torque increases as the coil current increases, and when $I = 1.8$ A, the torque reaches a value larger than 30 N m. The measured torque at 1.5 A coil current is close to the calculated value of 27 N m. To prevent the coil from being overheated, the maximum applied current here is 1.8 A (the electric resistance of the coil is 3.8 Ω and the power consumed by the coil is 12.3 W). The estimated torque of the knee joint during normal walking and standing up is smaller than 20 N m [4]. Therefore, the torque generated by the developed MR actuator is sufficient for normal activities of the knee joint.

4. Adaptive control of MR actuator

As the MR actuator exhibits nonlinear behavior, it is hard to precisely model it. In addition, system parameters may vary with time and environment. Hence, adaptive control is used to control the MR actuator here.

The system is described as

$$y_{k+1} = a y_k + b u_k \tag{6}$$

where y denotes the torque output, u is the control voltage, k is the sample index, and a and b are unknown gain parameters:

$$y_{r,k+1} = r_k \tag{7}$$

where y_r is the target torque output and r is the reference torque input. Let the control input be given by

$$u_k = c_k r_k + d_k y_k \tag{8}$$

where c and d are control gain parameters. There exist nominal parameter values:

$$c^* = \frac{1}{b} \qquad d^* = -\frac{a}{b} \tag{9}$$

such that the torque output matches the target torque output:

$$\begin{aligned} y_{k+1} &= a y_k + b(c_k r_k + d_k y_k) \\ &= (a + b d_k) y_k + b c_k r_k = r_k = y_{r,k+1} \end{aligned} \tag{10}$$

when $c_k = c^*$, $d_k = d^*$.

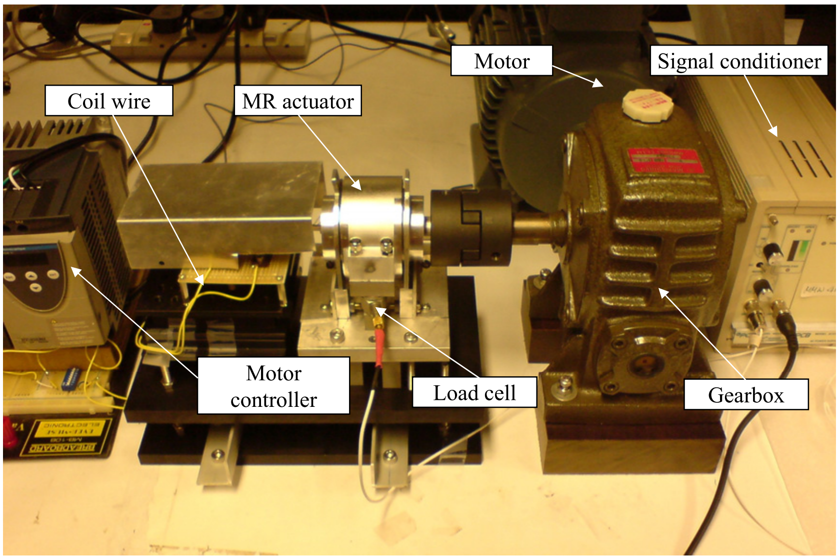


Figure 11. Photo of the experimental set-up.

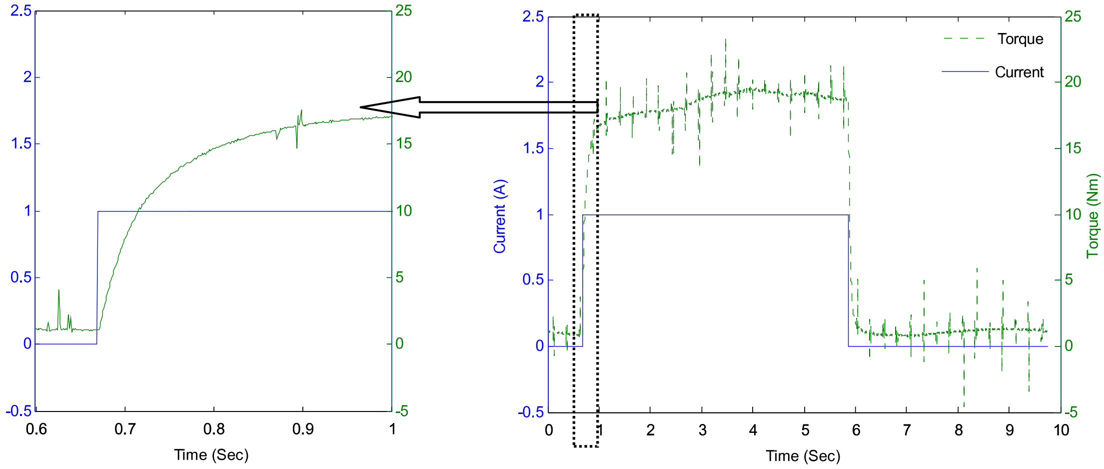


Figure 12. Torque of MR actuator under pulse current.

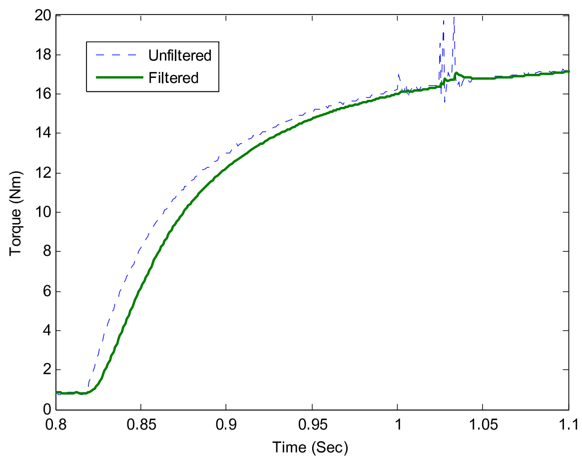


Figure 13. Comparison between filtered and unfiltered torque signals.

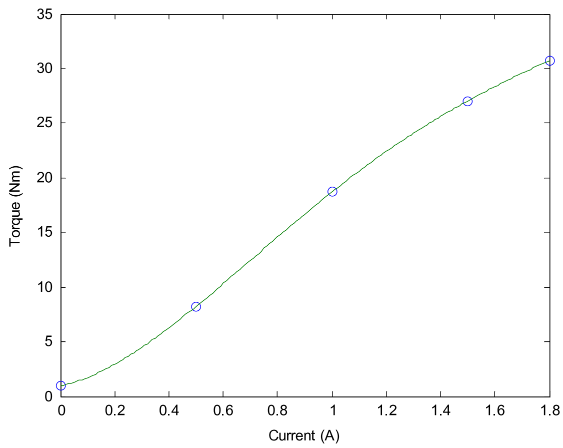


Figure 14. The relationship between applied current and output torque.

Define the output error:

$$e_k = y_k - y_{r,k} \tag{11}$$

and the parameter errors:

$$\Phi_{r,k} = c_k - c^*, \qquad \Phi_{y,k} = d_k - d^*. \tag{12}$$

Choose the following parameter update laws:

$$\begin{aligned} c_{k+1} &= c_k - g e_k r_k \\ d_{k+1} &= d_k - g e_k y_k, \qquad g > 0. \end{aligned} \tag{13}$$

Now the error formulation can be written as

$$\begin{aligned} \delta e_{k+1} &= e_{k+1} - e_k = -e_k + b(\Phi_{r,k} r_k + \Phi_{y,k} e_k + \Phi_{y,k} y_{r,k}) \\ \delta \Phi_{r,k+1} &= \Phi_{r,k+1} - \Phi_{r,k} = -g e_k r_k \\ \delta \Phi_{y,k+1} &= \Phi_{y,k+1} - \Phi_{y,k} = -g e_k^2 - g e_k y_{r,k}. \end{aligned} \tag{14}$$

Note that the right-hand sides only contain states (e_k, $\Phi_{r,k}$, $\Phi_{y,k}$) and exogenous signals ($r_k y_{r,k}$). Consider the Lyapunov function

$$v_k = \frac{e_k^2}{2} + \frac{b}{2g}(\Phi_{r,k}^2 + \Phi_{y,k}^2) \tag{15}$$

$$\delta v_{k+1} = v_{k+1} - v_k = -e_k^2 \leqslant 0. \tag{16}$$

Hence, δv_{k+1} is negative semi-definite and the parameter and output errors are upper bounded. It is demonstrated that, in the sense of Lyapunov, the adaptive system is stable. For all initial conditions, e_k, c_k and d_k are bounded. Since v_k is monotonically decreasing and lower bounded, $e_k \to 0$ as $k \to \infty$.

Experiments are carried out to test the torque tracking ability of the MR actuator under the adaptive control algorithm. The sampling rate is 1 kHz and the parameters are $c_0 = 7 \times 10^6$, $d_0 = -7 \times 10^6$ and $g = 0.9$. Figure 15 is the Simulink control model of the system. The DAC output of the dSPACE regulates the coil current and the ADC input of the dSPACE acquires the torque signal of the MR actuator.

Two kinds of signals are used as the reference torque: pulse and oscillatory signals. For the pulse signal, the amplitude is 10 N m and the pulse width is 4 s. For the

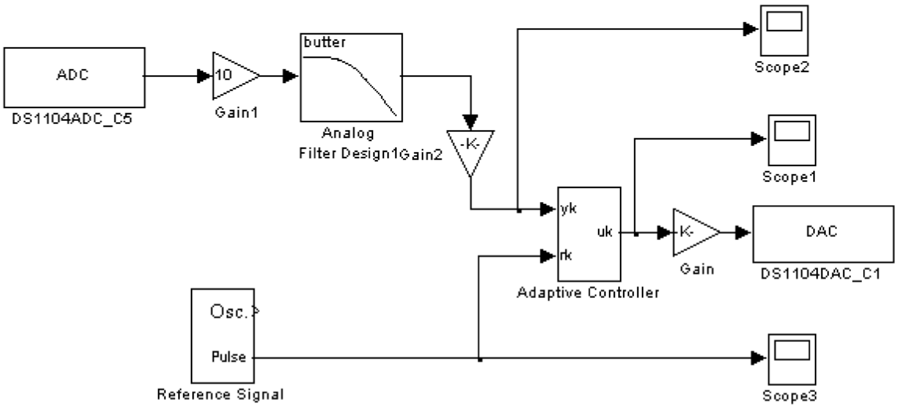


Figure 15. Simulink control model of the system.

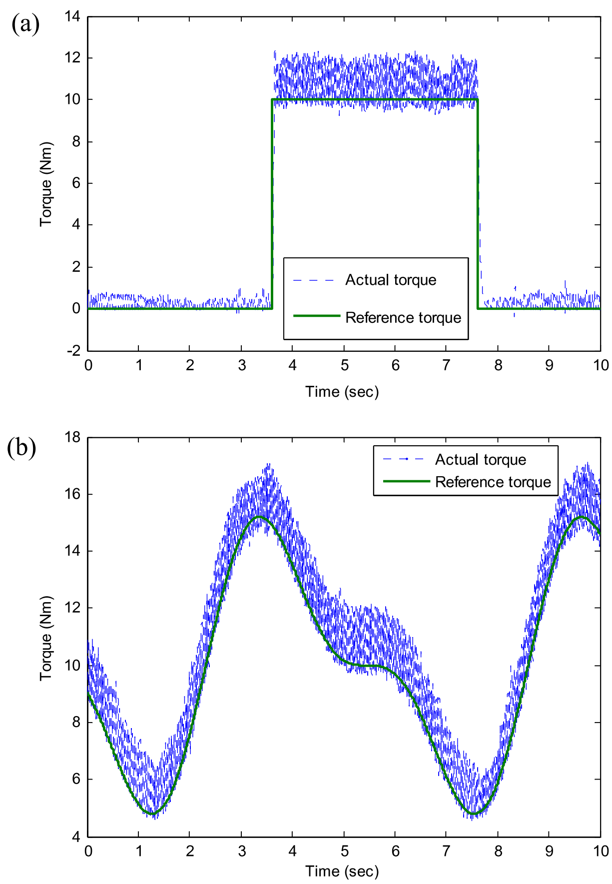


Figure 16. Torque tracking under adaptive control. (a) Pulse signal tracking. (b) Sinusoidal signal tracking.

oscillatory signal, the input reference torque is $y_r = 10 + 4\sin(2\pi t) + 2\sin(4\pi t)$. Figure 16 shows the experimental results. It shows that the system has good torque tracking ability under the adaptive control. However, oscillations happen during the experiments and the control algorithm should be improved to achieve better performance.

5. Improvement of control algorithm

The large oscillations for the torque output of the MR actuator are caused by large control gains. It is found that, even if we set the initial control gains very small, the parameter adaptation process would make the gains larger and larger. This phenomenon is caused by the saturation of the components used in the system. For example, when the control voltage output of dSPACE is 10 V (the upper limit of the dSPACE output), but the output torque is still smaller than the reference torque due to hysteresis or other reasons, the adaptation process will keep increasing the control gain to try to increase the output torque, which in fact has no effect on the output (due to saturation of dSPACE) while making the control gains drift to very large values. This phenomenon is called 'windup'. When the system is not saturated, it may then take a long time for the system to recover. To prevent the windup phenomena, an anti-windup strategy is added into the control:

(1) When the voltage control output of dSPACE $u \geqslant 10$, and the torque output of the MR actuator is smaller than the reference torque $y \leqslant y_r$, turn off the control gain adaptation.
(2) When the voltage control output of dSPACE $u \leqslant 0$, and the torque output of the MR actuator is larger than the reference torque $y \geqslant y_r$, turn off the control gain adaptation.

The above strategy can maintain the control parameters at reasonable values even if saturation or some other unexpected situation happened. However, small drift of the control parameters may occur when saturation happens due to hysteresis of the system. These small errors may accumulate eventually and can make the control parameters too large after a long period of operation. To guarantee long period stability, an auto-reset mechanism is added to the control.

Firstly, experiments are carried out to find the reasonable range of the control parameters. A detector is then added to monitor the control parameters. Once the parameters go beyond the 'right' range, the parameters are reset to their reasonable initial values.

The experimental results under the improved adaptive control shown above are shown in figure 17. The initial values are $c_0 = 6$ and $d_0 = -6$. It can be seen from figures 17(a) and (b) that the oscillations are eliminated and the controlled performances are greatly improved. Observing figure 17(c), there is a small change in every control parameter for each

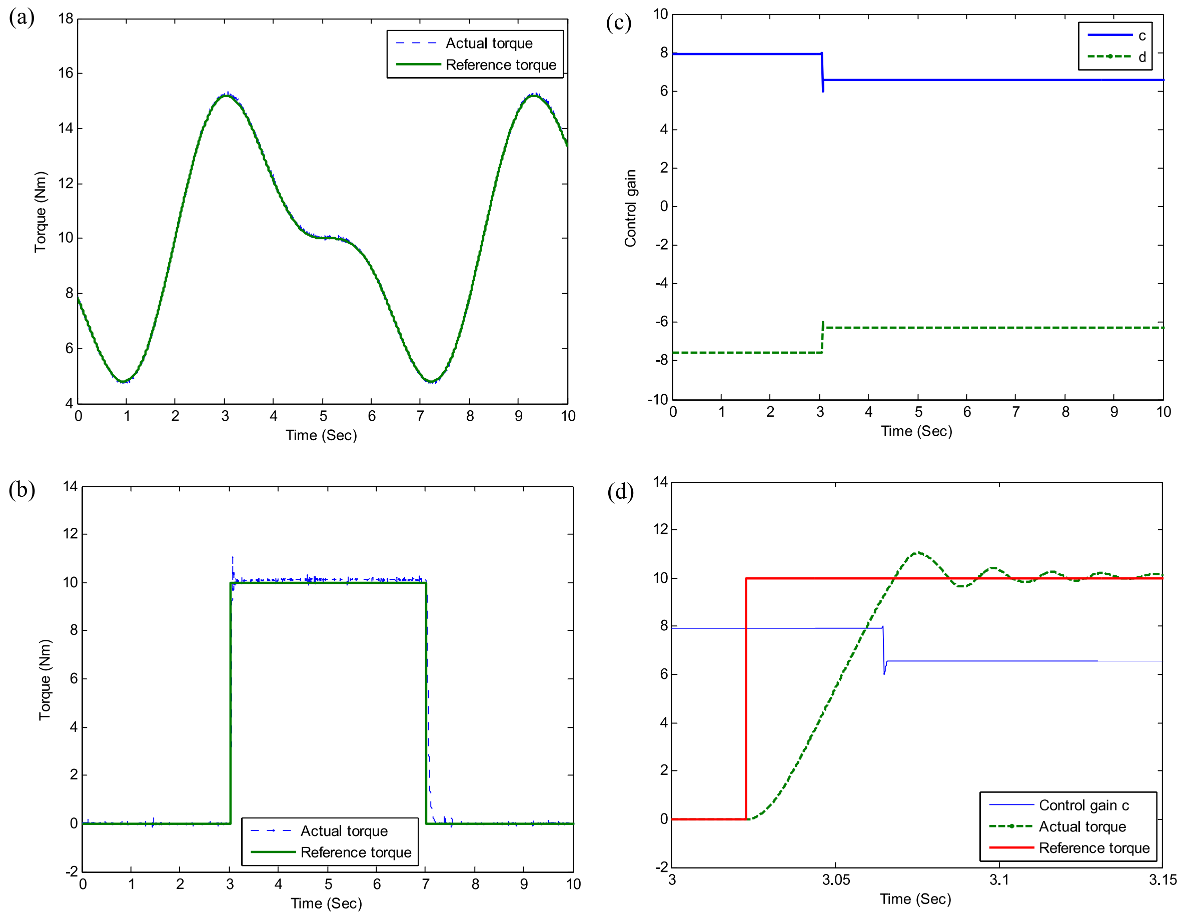


Figure 17. Torque tracking under improved adaptive control. (a) Sinusoidal signal tracking. (b) Pulse signal tracking. (c) Control parameters during pulse signal tracking. (d) Auto-reset process during pulse signal tracking.

step. These errors make the control gains larger and larger; once these values are equal to or larger than 8, they are reset to their initial values. It can be seen from figure 17(d) that this auto-reset mechanism would happen within a very short period of time and is stabilized very quickly. No disturbance to the output of the MR actuator is observed while the long-term stability is maintained. It can also be seen that the response time of the MR actuator under control is about 0.05 s, which is greatly improved compared to the open-loop response (refer to figure 12).

6. Conclusion

We developed an MR actuator that combines the advantages of controllable brake and clutch to work together with a DC motor for assistive knee braces. The prototype of the MR actuator was fabricated and tested. The experimental results showed that the MR actuator is able to provide sufficient torque needed for normal human activities. The torque can be further increased if needed. Adaptive control was proposed to achieve the desired torque output. The analysis and experimental results validate this algorithm. With anti-windup strategy and detect-auto-reset mechanism, the long-term control performance is guaranteed. This MR actuator together with a DC motor will be applied to an assistive knee brace in the near future. Besides the application to knee braces, the developed controllable MR actuator can also be applied to other devices such as leg exoskeletons, rehabilitation devices, etc.

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